

Worksheet Week 10 – Inheritance and Polymorphism

EC327 Introduction to Software Engineering – Spring 2026

1. Draw a diagram illustrating a base class (Animal) with two derived classes (Cat and Dog). Add a “Poodle”. Where would it go?
2. Give 2 examples of an “is-a relationship”.
3. Write a snippet of code to show “generic programming” using base and derived class objects.
4. In what order are constructors called between base and derived classes? Destructors? Why is it a good idea to make destructors virtual?

5. What is a virtual function? Give an example of a situation where you need a virtual function for correct behavior. What is a pure virtual function?

6. What is down-casting between objects? Up-casting? Why use dynamic casting with objects?